
Hotelogix

AI-Powered Hotel Management System

*A Final Year Project Report Submitted in Partial Fulfilment
of the Requirements for the Award of the Degree of
Bachelor of Business and Information Technology (BBIT)*

Submitted by: Abubakar Jamil
Registration: F22-BS-BBIT-1044
Batch: 22–26
Supervisor: Dr. Bilal
Department: Business & Information Technology
Institution: University of Okara
Academic Year: 2022–2026

August 5, 2026

Contents

Abstract	iii
1 Introduction	1
1.1 Background Information	1
1.2 Problem Statement	1
1.3 Overview of the Report	2
2 Purpose of the Project	4
2.1 Rationale	4
2.2 Goals	4
3 Significance of the Project	6
3.1 Impact	6
3.2 Value Addition	7
4 Objectives	8
5 Context Application	10
5.1 Relevance to Current Trends	10
5.2 Practical Application	11
6 Product Development Process	12
6.1 Idea Generation	12
6.2 Idea Screening	12
6.3 Concept Testing	13
6.4 Marketing Strategy	13
7 Website and Application Design	15
7.1 System Architecture	15
7.2 Design Process	16
7.3 Application Functionality	17
7.4 User Experience (UX)	18
8 Cost Benefit Analysis	19

8.1	Project Costs	20
8.2	Expected Benefits	20
8.3	Financial Analysis	21
9	Prototype and Its Manifestation	23
9.1	Prototype Development	23
9.2	Testing and Feedback	25
9.3	Improvements from Feedback	25
10	Commercialisation	27
10.1	Market Entry Strategy	27
10.2	Scalability	28
10.3	Sustainability	28
	Conclusion	30
	References	31

List of Tables

4.1	SMART Objectives for Hotelogix	9
6.1	Idea Screening Matrix	12
7.1	Hotelogix Technology Stack	16
8.1	Hotelogix Project Cost Breakdown	20
8.2	Net Present Value Calculation (3-Year Horizon)	22
9.1	Testing Summary	25

Abstract

Hotelogix is a full-stack, AI-powered hotel management system developed as a final year project for the Bachelor of Business and Information Technology (BBIT) programme. The system addresses the inefficiencies of traditional, manual hotel management by providing a unified digital platform that handles room browsing, online booking, dining reservations, loyalty deals, and administrative operations.

Two machine-learning subsystems are embedded in the platform. The first is an AI Room Recommendation Engine that uses a dual-model architecture – a Random Forest Classifier for user-room compatibility scoring and a Gradient Boosting Regressor for booking-likelihood prediction – achieving up to 79% classification accuracy and an R^2 of 0.60 for booking probability. The second is a Natural Language Processing (NLP) Chatbot trained on more than 555,000 labelled patterns across 64 intents, reaching 99.87% intent-classification accuracy.

The system is built on a React.js front end, a Node.js/Express back end, a Flask-based Python AI layer, and a MongoDB database. It is deployed on Vercel (client) and integrates Stripe for secure online payments. The project targets the Pakistani hotel sector, specifically Hotelogix properties in Okara, Lahore, Sheikhpura, and Multan.

Chapter 1

Introduction

1.1 Background Information

The hospitality industry in Pakistan is undergoing rapid digital transformation. Hotels of all sizes are under pressure to move away from paper-based reservation ledgers and fragmented software tools toward integrated, data-driven platforms that improve the guest experience while reducing operational overhead. Despite this pressure, many mid-tier and boutique hotel chains – particularly those operating in secondary cities such as Okara, Sheikhpura, and Multan – still rely on manual check-in registers, telephone-based bookings, and spreadsheet-driven administration.

The global hotel technology market is projected to exceed USD 11 billion by 2027, driven largely by property management systems (PMS), channel managers, and AI-powered personalisation engines. In Pakistan, the adoption curve lags behind the global average, creating both a gap and an opportunity. A locally-built, cost-effective management system that understands Pakistani pricing norms (PKR), user behaviour patterns, and regional languages has significant commercial and social value.

Hotelogix was conceived to fill exactly this gap: a modern, web-based hotel management platform built from first principles for the Pakistani market, enriched with artificial intelligence to deliver personalised recommendations and conversational customer support.

1.2 Problem Statement

The core problems that motivated this project are:

1. **Fragmented guest journey.** Prospective guests must call the hotel to inquire about rooms, visit in person to book, and interact with multiple staff members to manage a single stay. There is no unified self-service channel.
2. **Absence of personalisation.** Hotels present the same room catalogue to every

visitor regardless of their travel purpose, group size, budget, or past behaviour, resulting in low conversion rates and poor guest satisfaction.

3. **Manual administrative burden.** Booking management, cancellation processing, refund handling, and occupancy reporting are done manually, consuming significant staff time and introducing errors.
4. **No 24/7 support channel.** Customer queries outside business hours go unanswered, leading to lost bookings and negative reviews.
5. **Lack of data insights.** Hotel managers have no access to real-time analytics on room demand, user behaviour, or revenue performance, making strategic decisions purely intuition-based.

1.3 Overview of the Report

This report is structured as follows:

Chapter 1 – Introduction

Context, problem statement, and report overview.

Chapter 2 – Purpose of the Project

Rationale and goals of Hotelogix.

Chapter 3 – Significance of the Project

Impact on stakeholders and value addition.

Chapter 4 – Objectives

SMART objectives of the project.

Chapter 5 – Context Application

Alignment with current technology trends.

Chapter 6 – Product Development Process

Ideation, screening, concept testing, and marketing strategy.

Chapter 7 – Website and Application Design

Architecture, UI/UX, and functionality.

Chapter 8 – Cost Benefit Analysis

Financial feasibility, NPV, and ROI.

Chapter 9 – Prototype and Its Manifestation

Development, testing, and iteration.

Chapter 10 – Commercialisation

Market entry, scalability, and sustainability.

Conclusion

Summary and future directions.

Chapter 2

Purpose of the Project

2.1 Rationale

The motivation for building Hotelogix stems from three converging observations made during early research:

- A survey of hotel booking behaviour in Okara and Sheikhpura revealed that more than 70% of guests still book by telephone or walk-in, primarily because no online alternative exists for smaller local hotels.
- Existing PMS solutions available in Pakistan (e.g., Opera, Cloudbeds) are priced for large five-star properties and carry high monthly licensing fees that are prohibitive for independent or mid-tier hotel operators.
- Academic literature consistently shows that AI-driven personalisation increases hotel booking conversion rates by 15–30% compared to non-personalised platforms.

These observations confirmed that the market lacks an affordable, locally-tuned, AI-enriched hotel management platform. Building one as a BBIT project offered the opportunity to apply machine learning, full-stack web development, and business strategy skills in an integrated, real-world context.

2.2 Goals

The primary goals of the Hotelogix project are:

1. Build a fully functional, production-ready hotel management web application accessible from any device.
2. Implement an AI recommendation engine that personalises room suggestions for seven distinct user types using machine learning.
3. Deploy an NLP chatbot capable of answering customer queries in natural language with near-human accuracy (target: >95%).

4. Provide hotel administrators with a real-time analytics dashboard covering occupancy, revenue, booking trends, and AI model performance.
5. Integrate a secure online payment gateway (Stripe) to support cashless transactions.
6. Demonstrate commercial viability through a documented cost-benefit analysis and a market entry strategy.

Chapter 3

Significance of the Project

3.1 Impact

Hotel Operators

Hotel managers gain a centralised dashboard for bookings, cancellations, refunds, room inventory, dining reservations, and promotional deals. Tasks that previously required dedicated administrative staff can be handled autonomously by the system, reducing payroll costs and human error.

Guests

Guests benefit from a 24/7 self-service portal with AI-driven personalisation. Rather than browsing a static room catalogue, guests receive ranked recommendations tailored to their travel type (business, family, couple, solo, group, luxury, or budget), group size, season, and PKR budget. The integrated chatbot provides instant answers at any hour without requiring a staff member.

The Pakistani Hospitality Sector

By open-sourcing the architecture and targeting mid-tier properties, the project sets a precedent for affordable digital transformation in Pakistan’s secondary cities. Properties in Okara, Sheikhpura, and Multan – currently invisible to online travel agents (OTAs) – can compete with larger chains by offering a comparable digital experience.

Academic Contribution

The project demonstrates that a small BBIT team can design, train, and deploy production-grade ML models (Random Forest, Gradient Boosting, Logistic Regression with TF-IDF) within the constraints of an academic calendar, providing a replicable blueprint for future cohorts.

3.2 Value Addition

- **New AI tools for the sector:** A dual-model recommendation engine (compatibility + booking likelihood) is novel in the Pakistani mid-tier hotel context.
- **Synthetic data methodology:** The project documents a rigorous process for generating 200,000+ realistic training samples using domain-specific probability distributions, a method applicable to any data-scarce hospitality AI project.
- **Integrated analytics:** Real-time tracking of click-through rate (CTR), conversion rate, bounce rate, device breakdown, revenue, and time-of-day interaction patterns gives operators data-driven decision-making capability for the first time.
- **Competitor intelligence blocking:** The chatbot’s competitor-blocking feature protects brand integrity – a practically useful feature that also illustrates applied NLP intent engineering.

Chapter 4

Objectives

The following SMART objectives define the success criteria for Hotelogix:

SMART Criterion	Objective Detail
Specific	O1. Develop a full-stack web application with guest-facing and admin-facing modules. O2. Train and deploy an AI recommendation model covering 10 room types and 7 user types. O3. Build and deploy an NLP chatbot with ≥ 64 intents specific to Hotelogix properties. O4. Implement Stripe payment integration for secure on-line transactions. O5. Deliver a real-time admin analytics dashboard with ≥ 10 KPIs.
Measurable	• Recommendation engine compatibility accuracy $\geq 75\%$. • Chatbot intent accuracy $\geq 95\%$ on held-out test set. • All 30+ pages/routes functional with $< 3s$ average load time. • Zero critical security vulnerabilities in authentication flow.
Achievable	The project uses proven open-source frameworks (React, Node.js, Flask, scikit-learn, MongoDB) within the team's demonstrated skill set. Training data is synthetically generated to overcome real-data scarcity.
Relevant	All objectives align directly with digitising Hotelogix's operations, improving guest experience, and demonstrating applied AI within a business context – core pillars of the BBIT programme.

SMART Crite- rion	Objective Detail
Time-bound	- Months 1–2: Requirements, wireframes, database schema. - Months 3–4: Backend API and authentication. - Months 5–6: Frontend pages and AI model training. - Month 7: Integration, testing, and deployment. - Month 8: Report writing and viva preparation.

Table 4.1: SMART Objectives for Hotelogix

Chapter 5

Context Application

5.1 Relevance to Current Trends

Digital Transformation in Hospitality

Global hotel chains have invested heavily in digital-first guest experiences since 2020. Platforms like Booking.com and Airbnb have conditioned travellers to expect instant online booking, photo-rich room listings, and instant customer support. Hotelogix brings these capabilities to Pakistan’s independent hotel operators at a fraction of the cost.

Artificial Intelligence in Recommendation Systems

Collaborative filtering and content-based filtering are well-established in e-commerce (Amazon, Netflix). Applying similar principles to hotel rooms – using features such as user type, budget, group size, and stay duration – is an emerging research area. Hotelogix’s dual-model approach (compatibility classification + booking regression) reflects current best practices in the field.

Conversational AI and Chatbots

The global chatbot market is expected to reach USD 27 billion by 2030. NLP models based on TF-IDF and Naive Bayes or Logistic Regression remain highly effective for domain-specific intent classification where the vocabulary is controlled. Hotelogix’s 555,936-pattern training corpus and 99.87% accuracy demonstrate that academic projects can reach production-grade NLP quality.

Mobile-First and PWA Trends

With mobile internet penetration in Pakistan exceeding 50%, a mobile-responsive web application is not optional. Hotelogix’s React front end is fully responsive, tested

across desktop, tablet, and mobile viewports, and uses a proxy-based architecture compatible with Progressive Web App (PWA) conversion.

Cashless Payments

Pakistan’s State Bank digital payment initiatives (Raast, IBFT) and the rapid growth of debit card usage make online payment integration commercially viable. Stripe’s global infrastructure is integrated into Hotelogix to provide PCI-compliant card processing.

5.2 Practical Application

- A family planning a wedding-anniversary trip to Lahore uses the Smart Room Finder (`/smart-finder`) to input their budget (PKR 25,000/night), group size (2), and travel type (couple). The AI ranks all 10 room types and highlights the Executive Suite as a 91% match.
- A business traveller arriving in Multan at midnight types “do you have conference facilities?” into the chatbot. The system responds in under 200ms with accurate information about business amenities, entirely without staff involvement.
- A hotel manager in Okara logs into the admin dashboard (`/admin`) and views the previous week’s occupancy heatmap, top-booked room types, AI-driven revenue attribution, and a list of pending refund requests – all in a single screen.

Chapter 6

Product Development Process

6.1 Idea Generation

The project idea emerged from three parallel brainstorming sessions conducted in the first two weeks of the academic year:

1. **Problem-first brainstorm.** The team listed pain points observed during visits to hotels in Okara and Lahore: manual ledgers, no online booking, no personalisation, slow response to queries.
2. **Technology-first brainstorm.** Separately, the team listed technologies mastered in the programme: React, Node.js, Python/ML, MongoDB. A hotel management system provided a natural home for all of them.
3. **Market-gap analysis.** A search of Pakistani app stores and hotel websites confirmed the absence of an affordable, locally-built PMS with AI features.

The intersection of a clear market gap, a well-understood domain, and a matching technical skill set produced the Hotelogix concept.

6.2 Idea Screening

Three candidate projects competed for selection:

Project Idea	Domain Feasibility	Market Need	AI Scope	Score/15
Hotelogix PMS	Hospitality5/5	5/5	5/5	15
E-learning Platform	Education4/5	4/5	3/5	11
Ride-Hailing App	Transport2/5	4/5	3/5	9

Table 6.1: Idea Screening Matrix

Hotelogix scored highest on all three criteria and was selected unanimously.

6.3 Concept Testing

Before writing a single line of code, the concept was validated through:

- **Stakeholder interviews.** Three hotel front-desk managers in Lahore and Okara were interviewed. All confirmed that they would use an online booking system if it were affordable (under PKR 5,000/month) and easy to learn.
- **User survey (n=40).** A Google Form survey of potential guests found that 82.5% would prefer to book a hotel online if the interface were simple and payment were secure.
- **Low-fidelity wireframe testing.** Paper sketches of the home page, room listing, and booking flow were walked through with five participants. Feedback confirmed the navigation structure and surfaced the need for a visible chatbot button.
- **Supervisor review.** The academic supervisor reviewed the concept and recommended adding a cancellation and refund module, which was incorporated into the final scope.

6.4 Marketing Strategy

Target Audience

- **Primary:** Independent and mid-tier hotel operators in Pakistan’s secondary cities (Okara, Sheikhpura, Multan, Faisalabad) seeking affordable PMS solutions.
- **Secondary:** Hotel guests (25–50 years, smartphone users, middle and upper-middle income) who prefer digital self-service over telephone booking.

Marketing Mix (4Ps)

Product	A white-label SaaS hotel management platform with AI recommendation, NLP chatbot, online booking, dining reservations, deals management, Stripe payments, and a real-time admin dashboard. USP: the only AI-enriched PMS built specifically for Pakistan’s mid-tier market.
Price	Freemium model: free tier for single-property hotels (up to 20 rooms); PKR 4,999/month for the standard plan (unlimited rooms, AI features); PKR 9,999/month for the enterprise plan (multi-property, white-label branding, priority support). Competitive with no local equivalent.
Place	Delivered entirely via the web (hotelogix.vercel.app). No installation required. Mobile-responsive for guest use. Admin panel accessible

from any browser.

- Promotion**
- Social media campaigns on Facebook and Instagram targeting Pakistani hospitality groups.
 - LinkedIn outreach to hotel association members.
 - Demo videos on YouTube showing the AI recommendation and chatbot features.
 - Referral programme: existing hotel clients earn one free month per referral.
 - Academic press: presenting at BBIT conferences to attract investor attention.

Positioning

Hotelogix is positioned as “*The affordable, AI-powered hotel management system built for Pakistan*” – differentiating it from expensive international PMS solutions and from basic local booking portals that lack AI or admin capabilities.

Chapter 7

Website and Application Design

7.1 System Architecture

Hotelogix follows a three-tier, service-oriented architecture:

Tier 1 – Presentation Layer

React.js 18 SPA deployed on Vercel. Communicates with the back end exclusively via Axios HTTP calls. Mobile-responsive using custom CSS and the `mobile-responsive.css` stylesheet.

Tier 2 – Application Layer

Two independent servers:

- **Node.js / Express** (port 5000): REST API for all business logic – authentication (JWT), room management, bookings, cancellations, refunds, dining, deals, packages, and notifications.
- **Flask / Python** (ports 5001 & 5002): AI microservices. Port 5001 serves the NLP Chatbot API; port 5002 serves the Room Recommendation API.

Tier 3 – Data Layer

MongoDB Atlas (cloud) for all persistent data. Mongoose ODM used for schema validation and query building.

Technology Stack Summary

Layer	Technology	Purpose
Frontend	React.js 18, React Router v6	SPA, client-side routing
UI Libraries	react-toastify, react-datepicker	Notifications, date picking
Payments	Stripe, @stripe/react-stripe-js	Secure card processing
HTTP Client	Axios	API communication
Backend	Node.js, Express.js	REST API, business logic
AI Layer	Python 3, Flask, Flask-CORS	Recommendation & chatbot APIs
ML Libraries	scikit-learn, NumPy, Pickle	Model training & inference
NLP	NLTK, TF-IDF Vectoriser	Text preprocessing & vectorisation
Database	MongoDB Atlas, Mongoose	Data persistence
Auth	JWT, bcrypt	Secure authentication
Deployment	Vercel (frontend)	Production hosting
Version Control	Git / GitHub	Source code management

Table 7.1: Hotelogix Technology Stack

7.2 Design Process

Wireframing

Initial wireframes were drawn on paper and then recreated in Figma for 12 core screens: Home, Rooms Listing, Room Detail, Smart Room Finder, Booking Flow, My Bookings, Dining, Deals, Admin Dashboard, AI Analytics, Chatbot overlay, and Login/Register.

Prototyping

A clickable Figma prototype was shared with five test users. Key changes resulting from prototype feedback:

- The chatbot was moved from a sidebar to a floating bubble in the bottom-right corner to avoid obscuring content.
- The Smart Room Finder was given its own dedicated page (`/smart-finder`) rather than being embedded in the Rooms page.

- A cancellation flow with a confirmation modal (`CancellationModal.js`) was added after testers expressed concern about accidental cancellations.

UI Design

- **Colour palette:** Deep navy blue (`#003366`) as the primary brand colour, gold (`#C8A96E`) as the accent, white backgrounds for content areas.
- **Typography:** Inter / system-ui sans-serif stack for readability.
- **Component library:** Custom-built React components (Header, Footer, Loading spinner, ReviewForm, ReviewList, AdminNotifications, AIRecommendations, CancellationPolicy).

7.3 Application Functionality

The application comprises 30+ pages and routes, organised into four functional areas:

Guest-Facing Features

- **Room Browsing & Booking** (`/rooms`, `/rooms/:id`, `/book/:id`): Browse all available rooms with photos, amenities, pricing, and reviews. Book a room with date picker, guest count, and Stripe payment.
- **Smart Room Finder** (`/smart-finder`): AI-powered recommendation form. Guest inputs user type, season, day type, booking advance, stay duration, group size, budget (PKR), and view time. The recommendation API returns all 10 room types ranked by an overall score: $\text{Score} = (\text{Compatibility} \times 0.6) + (\text{Booking Likelihood} \times 0.4)$.
- **Dining** (`/dining`, `/restaurants/:id`, `/restaurants/:id/reserve`): Browse hotel restaurants and reserve a table.
- **Deals & Packages** (`/deals`, `/packages`): Browse promotional deals, redeem loyalty offers, and book bundled packages.
- **My Bookings** (`/my-bookings`): View booking history with status, dates, and invoice details.
- **Cancellation & Refund** (`/cancel-booking/:id`): Self-service cancellation with policy display and automated refund initiation.
- **AI Chatbot** (global, all pages): Floating NLP chatbot powered by the 200K+ pattern model. Answers queries about rooms, pricing, amenities, cities, booking procedures, and contact details. Blocks competitor queries.

- **Profile** (/profile): Manage personal information and view loyalty status.
- **Static Pages**: About, Contact, FAQ, Privacy Policy, Terms of Service, Sitemap.

Admin-Facing Features

- **Admin Dashboard** (/admin): Manage all bookings, rooms, dining, deals, and packages. Process cancellations and refunds.
- **Notifications** (/admin/notifications): Real-time alerts for new bookings, cancellation requests, and refund approvals.
- **AI Analytics** (/ai-analytics): Live dashboard showing recommendation engine KPIs – total predictions, CTR, conversion rate, bounce rate, average session time, device breakdown, time-of-day interaction heatmap, room-type distribution, revenue attribution, and AI-driven revenue contribution.
- **Refund Processing** (/admin/refund/:id): Review and approve or reject refund requests with audit trail.

7.4 User Experience (UX)

- **Scroll-to-top**: A ScrollToTop component ensures every page navigation begins at the top of the viewport.
- **Loading states**: A Loading component with spinner is displayed during all async API calls to prevent blank screens.
- **Toast notifications**: react-toastify provides non-intrusive success/error/info messages.
- **Route protection**: Authenticated routes (/bookings, /profile, /admin/*) are only rendered when a valid JWT is present; admin routes additionally check for an admin email pattern.
- **Header visibility**: The header and footer are hidden on the Login and Register pages to reduce distraction during authentication.
- **Mobile responsiveness**: All layouts use fluid grids and media queries. The application is tested at 375px (iPhone SE), 768px (tablet), and 1280px (desktop) breakpoints.
- **Accessibility**: Semantic HTML5 elements (<main>, <nav>, <section>), ARIA labels on interactive controls, and sufficient colour contrast ratios are used throughout.

Chapter 8

Cost Benefit Analysis

8.1 Project Costs

Category	Item	One-Time (PKR)	Annual (PKR)
3*Development	Developer time (8 months, team of 2)	0 (academic)	–
	Software licences (all open-source)	0	0
	Design tools (Figma free tier)	0	0
3*Infrastructure	Vercel Pro hosting (frontend)	–	36,000
	MongoDB Atlas M10 cluster	–	60,000
	Domain name (.com, 1 year)	3,500	3,500
2*AI / Python	Google Colab Pro (model training)	6,000	6,000
	Flask VPS server (2 vCPU, 4 GB RAM)	–	48,000
2*Payments	Stripe setup (free)	0	0
	Stripe transaction fee (2.9% + \$0.30)	–	Variable
Marketing	Social media ads (launch campaign)	25,000	60,000
2*Miscellaneous	Testing devices (mobile)	10,000	0
	Contingency (10%)	4,450	21,350
Total		48,950	234,850

Table 8.1: Hotelogix Project Cost Breakdown

8.2 Expected Benefits

Revenue Streams

- **SaaS subscriptions:** PKR 4,999/month (standard) per hotel property. Target: 20 properties in Year 1 $\Rightarrow$ PKR 1,199,760/year.
- **Enterprise licences:** PKR 9,999/month. Target: 5 enterprise clients in Year 1 $\Rightarrow$ PKR 599,940/year.
- **Transaction commission:** 1% of each booking processed through the platform. Assuming avg. booking PKR 12,000 and 500 bookings/month across all clients $\Rightarrow$ PKR 720,000/year.
- **Total projected Year 1 revenue:** PKR 2,519,700.

Cost Savings for Hotel Operators

A typical mid-tier hotel using a manual system spends approximately PKR 30,000–50,000/month on administrative staff for booking management. Hotelogix reduces this to PKR 5,000–10,000/month (one part-time staff member for exception handling), a saving of PKR 240,000–480,000/year per property.

Intangible Benefits

- Improved guest satisfaction through 24/7 chatbot support and personalised recommendations.
- Increased booking conversion rate (estimated +15–20%) due to AI recommendations.
- Enhanced brand visibility through digital presence previously absent.
- Real-time data for strategic decisions (pricing, promotions, staffing).

8.3 Financial Analysis

Net Present Value (NPV)

Using a discount rate $r = 12\%$ over 3 years:

$$NPV = -C_0 + \sum_{t=1}^3 \frac{B_t - C_t}{(1 + r)^t}$$

Year	Revenue (PKR)	Costs (PKR)	Net Cash Flow	PV @ 12%
0 (Setup)	0	48,950	−48,950	−48,950
1	2,519,700	234,850	2,284,850	2,039,152
2	3,779,550	280,000	3,499,550	2,790,238
3	5,039,400	320,000	4,719,400	3,358,924
			NPV	PKR 8,139,364

Table 8.2: Net Present Value Calculation (3-Year Horizon)

A positive NPV of **PKR 8.14 million** confirms the project is financially viable.

Return on Investment (ROI)

$$ROI = \frac{\text{Net Benefit}}{\text{Total Investment}} \times 100 = \frac{2,284,850}{283,800} \times 100 \approx \mathbf{805\%}$$

The exceptional ROI reflects the near-zero marginal cost of software replication once the platform is built.

Payback Period

At a Year 1 monthly net cash flow of approximately PKR 190,404, the initial investment of PKR 48,950 is recovered in under **one month** of operations.

Chapter 9

Prototype and Its Manifestation

9.1 Prototype Development

Phase 1 – Backend API (Weeks 1–6)

The Node.js/Express REST API was built first, establishing all data models (User, Room, Booking, Dining, Deal, Package, Notification) in MongoDB via Mongoose. JWT-based authentication with bcrypt password hashing was implemented and tested using Postman. All endpoints were verified before any frontend work began.

Phase 2 – AI Model Training (Weeks 4–8, parallel)

Recommendation Engine:

- 100,000 synthetic training samples generated using domain-specific probability distributions: exponential for `booking_advance`, gamma for `stay_duration`, and Gaussian noise for variability.
- Compatibility matrices encoded domain knowledge: families prefer Family Suites (95% probability), business travellers prefer Business Rooms (95%).
- Models trained: Random Forest Classifier (compatibility) and Gradient Boosting Regressor (booking likelihood).
- Final metrics: compatibility accuracy = **79.0%**, booking $R^2 = \mathbf{0.597}$, booking RMSE = **0.111**.
- Features scaled with `StandardScaler` (mean=0, variance=1) to handle the wide range of PKR budget values (2,000–150,000) versus group size (1–10).
- Artefacts saved: `compatibility_model.pkl`, `booking_model.pkl`, `scaler.pkl`, `label_encoders.pkl`.

NLP Chatbot:

- Three model iterations: basic (52 intents), mega (64 intents, 92,436 patterns), and

ultra/200K (64 intents, 555,936 patterns).

- Pipeline: NLTK tokenisation + lemmatisation → TF-IDF Vectoriser (`max_features=5000`, `ngram_range=(1,3)`) → Logistic Regression classifier.
- Final accuracy on held-out test set: **99.87%**.
- Intent coverage: room types, pricing, booking procedure, amenities, city-specific questions (Okara, Lahore, Sheikhpura, Multan), contact details, FAQs, and 15+ competitor-blocking intents.
- Artefacts saved: `hotelogix_200k_model.pkl`, `hotelogix_200k_intents.pkl`.

Phase 3 – Frontend Development (Weeks 6–14)

React.js SPA built with React Router v6. Components developed iteratively:

- Shared: Header, Footer, Loading, ScrollToTop, Chatbot.
- Guest pages: Home, Rooms, RoomView, BookRoom, SmartRoomFinder, Dining, Deals, Packages, MyBookings, CancelBooking, Profile.
- Admin pages: AdminDashboard, AIAalytics, Notifications, ProcessRefund.
- Static pages: About, Contact, FAQ, Privacy, Terms, Sitemap.

Axios instance configured in `src/api/axios.js` with base URL proxied to `http://localhost:5000`.

Phase 4 – Integration and Deployment (Weeks 14–16)

- Flask CORS enabled on both AI APIs to allow cross-origin requests from the React frontend.
- Stripe Elements integrated using `@stripe/react-stripe-js` and `@stripe/stripe-js` v8.2.
- Frontend built with `CI=false` flag and deployed to Vercel. Production environment variables stored in `.vercel/.env.production.local`.
- Vercel project configuration stored in `client/.vercel/project.json`.

9.2 Testing and Feedback

Test Type	Tool / Method	Key Findings
Unit Testing	Manual Postman	All 25+ API endpoints returned correct status codes and payloads.
ML Model Testing	scikit-learn train/test split (80/20)	Recommendation accuracy 79%, chatbot 99.87%.
Integration Test	Axios + Live Flask APIs	Chatbot responses returned in <200ms; recommendation scores in <300ms.
UAT	5 student volunteers	4/5 completed a booking without assistance; chatbot correctly answered 18/20 queries.
Cross-browser	Chrome, Firefox, Edge	No layout breakage; all routes functional.
Mobile	iPhone 12, Samsung A52	Responsive layouts confirmed; chatbot bubble visible and functional.
Payment	Stripe test mode	All card scenarios (success, decline, 3DS) handled correctly.

Table 9.1: Testing Summary

9.3 Improvements from Feedback

- UAT testers found the booking confirmation page lacked a clear summary. A booking-confirmation card with room details, dates, total price, and a booking reference was added.
- One tester attempted to cancel a booking that was already checked in; a server-side guard and a clear error message were added.
- The chatbot initially gave verbose responses on mobile, truncating the last sentence. Response length was capped at 200 characters for intents flagged as “brief”.

- Low-confidence chatbot queries (confidence < 0.30) were logged and used to add 12 new training patterns in the final model iteration.
- The AI Analytics page initially loaded all stats at once, causing a 2-second blank screen. A skeleton loader was added and the API was split into two lighter endpoints.

Chapter 10

Commercialisation

10.1 Market Entry Strategy

Phase 1 – Pilot Launch (Months 1–3)

Deploy Hotelogix in one real hotel property in Okara under a free pilot agreement. Collect real booking data, monitor system performance, and gather testimonials. Use the pilot to produce a case-study video and a before/after operational efficiency report.

Phase 2 – Regional Launch (Months 4–9)

Expand to 10–15 hotel properties across Okara, Sheikhpura, Lahore, and Multan using the standard PKR 4,999/month subscription tier. Drive acquisition through:

- Direct sales calls to hotel managers using LinkedIn and hospitality association directories.
- Demonstration webinars showing the AI recommendation and chatbot features live.
- A referral programme offering one free month for each new client introduced.

Phase 3 – National Scale (Year 2+)

Expand to 50+ properties nationwide. Introduce the enterprise plan (PKR 9,999/month) for hotel chains and multi-property operators. Explore partnerships with Pakistan's OTAs (Jovago, Wego) for listing integration.

Distribution Channels

1. **Direct SaaS:** Hotel operators sign up on the Hotelogix website, configure their property, and go live within 24 hours.

2. **Reseller network:** IT consultants and hospitality consultants earn a 20% commission on first-year subscription revenue for each client they onboard.
3. **OTA integration:** API-level integration with booking aggregators to funnel external bookings into the Hotelogix dashboard.

10.2 Scalability

Technical scalability:

- The React frontend on Vercel scales automatically with zero infrastructure management.
- MongoDB Atlas supports horizontal sharding for database scalability beyond single-cluster limits.
- The Flask AI microservices can be containerised (Docker) and orchestrated with Kubernetes for horizontal scaling under high inference load.
- The Node.js API can be deployed as serverless functions (Vercel Serverless or AWS Lambda) to scale to zero and burst on demand.

Business scalability:

- The white-label architecture allows any hotel to customise branding (logo, colour scheme, domain) without code changes.
- Multi-tenancy support means a single deployment serves unlimited hotel properties with data isolation per tenant.
- The AI models can be retrained automatically on a weekly/monthly schedule using real booking data accumulated from all properties, continuously improving recommendation accuracy.

10.3 Sustainability

Revenue Model Longevity

The SaaS subscription model generates recurring monthly revenue, providing financial stability. Churn risk is low because switching to a competitor PMS requires staff retraining and data migration – high switching costs that favour retention.

Technical Maintenance

- Dependency updates scheduled quarterly using `npm audit` and `pip check`.

- Model performance monitored via the `/stats` endpoint; automated retraining triggered if compatibility accuracy drops below 75%.
- MongoDB Atlas automated backups (daily snapshots, 7-day retention) protect against data loss.

Long-Term Product Roadmap

1. **Year 2:** Urdu language support for the chatbot using multilingual NLP models (mBERT or IndicBERT).
2. **Year 2:** Mobile app (React Native) for guests and hotel staff.
3. **Year 3:** Deep learning recommendation upgrade using neural collaborative filtering on accumulated real booking data.
4. **Year 3:** Dynamic pricing engine that adjusts room rates based on demand prediction, competitor pricing, and season.
5. **Year 4:** Regional expansion to Bangladesh and Sri Lanka, localising the platform for those markets.

Environmental and Social Sustainability

- Eliminating paper-based booking registers and printed invoices reduces paper waste across all client properties.
- Cloud hosting on Vercel and MongoDB Atlas (both with renewable energy commitments) minimises the carbon footprint relative to on-premise servers.
- By enabling smaller, locally-owned hotels to compete digitally, the platform supports local entrepreneurship and employment in Pakistan's secondary cities.

Conclusion

Hotelogix represents a complete, production-grade solution to a real problem in Pakistan’s hospitality sector. Starting from a clearly articulated problem statement – the absence of an affordable, AI-enriched digital management platform for mid-tier hotels – the project systematically progressed through ideation, concept validation, iterative development, testing, and deployment.

The two AI subsystems are the defining technical achievements of the project. The dual-model recommendation engine (Random Forest + Gradient Boosting, 79% accuracy, $R^2 = 0.60$) delivers genuinely personalised room rankings across 10 room types and 7 user profiles. The NLP chatbot (Logistic Regression + TF-IDF, 555,936 training patterns, 99.87% accuracy) provides hotel-specific conversational support at scale. Together they create a guest experience that rivals systems deployed by five-star international chains, built entirely with open-source tools.

The financial analysis confirms strong commercial viability: a positive NPV of PKR 8.14 million over three years, an ROI of 805%, and a payback period of under one month. The phased commercialisation strategy – pilot, regional rollout, national scale – provides a realistic path from academic prototype to live business.

Future work will focus on transitioning from synthetic to real booking data for model retraining, adding Urdu language support, launching a React Native mobile app, and exploring dynamic pricing. Hotelogix is not merely a final year project; it is the foundation of a deployable product.

References

- [1] Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), 5–32.
- [2] Friedman, J. H. (2001). Greedy Function Approximation: A Gradient Boosting Machine. *Annals of Statistics*, 29(5), 1189–1232.
- [3] Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- [4] Salton, G., & Buckley, C. (1988). Term-weighting approaches in automatic text retrieval. *Information Processing & Management*, 24(5), 513–523.
- [5] Bird, S., Klein, E., & Loper, E. (2009). *Natural Language Processing with Python*. O'Reilly Media.
- [6] React Documentation. (2024). React – The library for web and native user interfaces. <https://react.dev>
- [7] MongoDB Documentation. (2024). MongoDB Manual. <https://www.mongodb.com/docs/manual/>
- [8] Stripe Documentation. (2024). Stripe API Reference. <https://stripe.com/docs/api>
- [9] Vercel Documentation. (2024). Vercel Platform Overview. <https://vercel.com/docs>
- [10] Pakistan Telecommunication Authority. (2024). *Annual Report 2023–24*. PTA, Islamabad.